

VISUALIZING FRIENDSHIPS



A Facebook intern visually maps the 500 million friendships that exist on the social networking site. Check out the visual to see how geography and political borders influence where people live relative to their friends.

<http://on.fb.me/fcG8Ng>

CULTUROMICS

Digital tools such as ngrams. googlelabs.com are propagating this branch of study and research. With the tool you can chart the usage of words historically within Google's nearly 5.2 million digitized books right back to the 1800s. Some very interesting cultural inferences can be drawn from it. Highly addictive.

<http://ngrams.googlelabs.com/>

WHEN COMPUTERS GO WRONG

A look at ten IT disasters that highlight the precariously fickle nature of computers. The screwups highlighted in this article may have caused lasting damage to the finances or reputations of those involved, but thankfully nobody was physically harmed in the making of this list.

<http://bit.ly/fil18Vj>

way around this problem that could both benefit Windows XP users and ease the transition for everyone to the larger block sizes: Hard drives would report their blocks as only being 512 bytes in size, so any OS that didn't understand the larger sizes could just "see" each 4KB block as eight 512-byte blocks. Windows XP, then, would just need to "align" its sectors properly; that was hardly an insurmountable problem and, indeed, Western Digital has designed software to compensate.

Unfortunately, even 4KB sectors couldn't guarantee drives of infinite capacity. Because of the limitations of the BIOS and 32-bit MBR partition in the earliest years of computing, hard drive capacities had a firm upper ceiling of 232 logical blocks. If each of those blocks was 512 bytes in size, that meant that MBR drives could store no more than 2,199,023,255,552 bytes, or 2.19TB.

More Problems and More Solutions

For computers to recognize drives any larger than 2.19TB, the MBR and the BIOS would have to be replaced. The successor to the MBR is the GUID Partition Table (GPT), which offers 64-bit block addressing, and thus (when 4KB blocks are utilized) a maximum storage size of 9.4 zettabytes (or 9.4 trillion gigabytes). What's supplanting BIOS, which can't read GPT, is the Unified Extensible Firmware Interface (UEFI), which is built on CPU-independent architecture and drivers and offers more flexibility and features outside of the operating system. Because Windows XP has no support whatsoever for GPT and UEFI, no system running it can natively use any drive with a capacity over 2.19TB. Even if you have a supported OS (Windows Vista, Windows 7, and most flavors of Linux) that recognizes

GPT, you won't be able to boot to a drive of that size unless you also have a motherboard running UEFI—something that, as of this writing in late 2010, very few do. All current Intel boards support UEFI, but almost no other major manufacturer has yet followed suit. So, under most circumstances, if you don't have a UEFI motherboard (and you probably don't), you'll have to use your extra-large hard drive for storage only. (There are worse things.)

Finally, your system's SATA controller must also be designed to recognize 4KB blocks. This isn't necessarily a big deal: As we discovered when we reviewed Western Digital's new 3TB Caviar Green hard drive, the company is including with all its above-2.19TB drives a PCI Express x1 Host Bus Adapter that lets Windows use a known driver to communicate with the drive.

What Does the Future Hold?

Right now, there aren't a lot of compelling reasons for most home users to worry about drives larger than 2.19TB. Factor in the compatibility problems, and you're investing more money and time in a product than you may necessarily want to. Right now, the less-expensive, faster, easier-to-understand-and-configure storage is the better choice all the way around.

This isn't to say this will be the case forever—or even for very long. As of this writing, Asus has just introduced a software app called Disk Unlocker that will let users partition storage space above 2.19TB as a virtual drive. But it has a bunch of restrictions, not the least of which is that it only works with Asus motherboards. It's an imperfect fix, but it shows what's possible.

It also shows how far we still have to go. In many cases, the software irritations have



already been soothed: 64-bit operating systems now make up a vast percentage of the market; and Windows XP is slowly fading away, and with it MBR. That leaves the current bottleneck as hardware: The only sure-fire way to deal with these problems once and for all is for motherboard manufacturers to universally implement UEFI and use SATA controllers that recognize 4KB blocks, two things that haven't happened yet. Hard drive makers agreeing on 2011 as the first official year in which 4KB blocks will be the rule and 512-byte blocks the exception is a good first step, but it can't be the last one. We have no doubt that WD and other hard drive companies will continue pushing the boundaries of storage capacity. But right now, too few users are properly equipped to follow where they're leading.